

Madame

TD 09



Exo 1:

ligne l $x^l = (x_1^l, x_2^l, x_3^l)$ sortie y^l

On veut $\min \sum_{l=1}^n |y^l - C_v(x_1^l, x_2^l, x_3^l)|$

On cherche la capacité $v: \{ \emptyset: 0, \{1\}: v_1, \{2\}: v_2, \{3\}: v_3, \{1,2\}: v_{12}, \dots, \{1,2,3\}: v_{123} \}$

1. $\min \sum_{l=1}^{10} (u_l + |w_l|)$ linéarité de la val° abs° Régression LAD

$$\begin{aligned} u^l - w^l &= y^l - C_v(x_1^l, x_2^l, x_3^l) \quad \forall l \\ \text{s.c.} \quad v_A &\leq v_B \quad \forall A \leq B \leq \{1,2,3\} \\ v_{\{1,2,3\}} &= 1 \end{aligned}$$

$$u_1 - w_1 = 0.3101 - v_{12}$$

$$u_2 - w_2 = 0.4285 - v_{13}$$

$$u_3 - w_3 = 0.5055 - v_{23}$$

$$u_4 - w_4 = 0.5335 - 0.5$$

$$u_5 - w_5 = 0.3823 - (0.25 + 0.25 \cdot v_{12} + 0.25 \cdot v_{13})$$

$$u_6 - w_6 = 0.4560 - (0.25 + 0.25 \cdot v_{12} + 0.25 \cdot v_{13})$$

+ petit = $v_{\{1,2,3\}}$ puis + petit + petit = $v_{\{1,2,3\}}$ (on relie la diff° correspondante colonne)
et ainsi de suite

$$\begin{aligned} v_1 &\leq v_{12} & v_2 &\leq v_{13} & v_{12} &\leq 1 \\ v_1 &\leq v_{13} & v_3 &\leq v_{23} & v_{23} &\leq 1 \\ v_2 &\leq v_{12} & & & v_{13} &\leq 1 \\ v_2 &\leq v_{23} & v_3 &\geq 0 & & \forall s \end{aligned}$$

À linéariser

$$\lambda \sum_{B \in N(S) \geq 2} (\alpha_B + \beta_B)$$

2. $\min \sum_{l=1}^{10} (u_l + |w_l|) + \lambda \sum_{B \in N(S) \geq 2} |m_B|$ Möbius Régression LAD-LASSO

$$\begin{aligned} u^l - w^l &= y^l - \sum_{B \in N(S) \geq 2} m_B \min_{i \in B} \{x_i\} \quad \forall l \\ \text{s.c.} \quad \sum_{B \in S, B \ni i} m_B &\geq 0 & v(S) &\leq v(S \setminus \{i\}) & \forall S, i \in S \\ \alpha_B + \beta_B &= m_B & \forall B \in N(S) &\geq 1 \end{aligned}$$

$$u^1 - w^1 = 0.3101 - (m_1 + m_2 + m_{12})$$

$$\begin{aligned} u^5 - w^5 &= 0.3823 - (0.75 m_1 + 0.5 m_2 + 0.25 m_3 \\ &\quad + 0.5 m_{12} + 0.25 m_{13} \\ &\quad + 0.25 m_{13} + 0.25) \end{aligned}$$

$$m_1 + m_{12} + m_{123} \geq 0$$

$$m_1 + m_{12} \geq 0$$

min° à chaque comp! !

3. On a fait sauter des coeffs pas pertinents, sans perte de précision. On peut expliquer le rés° avec slt° 4 var°.

Exo 3:

$$\begin{aligned} 1. \quad C_v(z^2) &= 1 + 5v_2 \\ C_v(z^4) &= 0 + 10v_2 \\ 1 + 5v_2 &\geq 0 + 10v_2 \\ 1 &\geq 5v_2 \\ v_2 &\leq \frac{1}{5} = 0,2 \end{aligned}$$

$$\begin{aligned} C_v(z^4) &= 1 + 2v_2 \\ C_v(z^5) &= -1 + 6v_2 \\ 1 + 2v_2 &\geq -1 + 6v_2 \\ 2 &\geq 4v_2 \\ \frac{1}{2} &\geq v_2 \end{aligned}$$

$$C_v(z^3) = 2 + v_2$$

Rejet de choisir z^3 plutôt que z^2

$$PMR(z^3, z^2, v) = \max_{v \in V} \left\{ \underbrace{1 + 5v_2 - 2 - v_2}_{\substack{-1 + 4v_2 \\ \leq 1}} \right\} \leq 0$$

$$PMR(z^3, z^4, v) = \max \left\{ \underbrace{1 + 2v_2 - 2 - v_2}_{\substack{-1 + 2v_2 - v_1 \\ \leq 1}} \right\} \leq 0$$